

Simple Interest

Conversion PPT

Simple Interest → every year same

1) Simple Interest = ?

$$\# \boxed{P \times R \times T = SI}$$

Q On a sum of 10,000 at the rate of 10% for 2 years what will be the amount?

$$\boxed{A = P + SI} \quad \text{Ans} = 12000$$

2) ~~Unit~~ Principle = ?

(i) Multiply $R\% \times T$

$$\# (ii) R\% \times T = SI$$

figure out the values of 1% then figure out the 100% value that will be your principle

Q On a certain sum at the rate of 10% for 5 years the ~~pr~~ SI was 10,000 figure out the sum?

Ans =

$$10\% \quad 5 = 10,000$$

$$\frac{10}{100} \quad 50\% = 10,000$$

$$1\% = 200$$

$$100\% = 20,000$$

3) $R\% = ?$

Simple Interest for 1 year $\times 100$ principle

Q A sum of 10,000 amount to 12,000 in 2 years figure out for rate of int

$$\frac{10000}{100000} \times 10000$$

$$R\% = 10\%$$

$$P = 10\%$$

4) Time = ?

(i) Make a fraction of Rate of Interest if 10% $= 10\% = \frac{1}{10}$

$P = \text{unit}$ | 1 year $SI = 1 \text{ unit}$

(ii) Make your unit value equal to the given value

Q A sum of 10,000 at the rate of 10% in how many years it will amount to 12,000
 $= 2 \text{ years}$

Unit principle = given principle

$$10 = 10,000$$

(iii) Given SI

SI for 1 year

$\propto$ Time

$$\frac{2000}{10000} = 2 \text{ years}$$

Q In how many years amount of 450 yield 81 at the rate of int 4.5

$$4.5\% = \frac{4.5}{100} = \frac{45}{1000}$$

$\frac{9}{1000}$ p = 20 unit
1 year S1 = 9 unit

unit = actual value

$$\begin{array}{rcl} 2000 & - & 450 \\ 4 & - & 9 \\ 1 & & 9.25 \end{array}$$

1 unit = 2.25

$$\frac{81-9}{9 \times 2.25}$$

(4 year)



Compound Interest

every year different

1) CI = ?

Q = On 6000 at the rate of 10%
Per annum of what will be the amount two years?

[R%]

(i) fraction of R%

$$10\% = \frac{1}{10}$$

$$\begin{array}{rcl} P + I & & A \\ 10 + 1 & = & 11 \\ (10)^2 & = & (11)^2 \\ 100 & = & 121 \end{array}$$

P = 100 unit
A = 121 unit

$$CI = A - P = 21 \text{ unit}$$

$$\begin{array}{rcl} 100 & - & 1000 \\ 1 & = & 10 \end{array}$$

$$121 = 1210 \quad 121 \times 10$$

$$= 1210$$

(Ans)

Three year (10)³

$$10\% = \frac{1}{10}$$

$$\begin{array}{rcl} P & & A \\ (10)^3 & & (11)^3 \\ 1000 & & 1331 \end{array}$$

$$1000 = 1000$$

$$1331 = 1331$$

Q what will be the amount on 8000 at the rate 5% for three years on a compound interest =

$$5\% = \frac{5}{100} = \frac{1}{20}$$

$$\begin{array}{rcl} P & & A \\ (20)^3 & & (21)^3 \\ 8000 & & 9261 \end{array}$$

$$8000 = 8000$$

$$9261 = 9261$$

Compound interest

2) R% = ?

Q If a sum of 24000 amount to 27783 in a 3 years figure out the rate of interest.

Q In 2 years a sum of 2000 amount to 2880 figure out the rate of interest?

P	A
2/2000	2/2880
100	144
(10)	(12)
$\frac{1}{5}$	$\frac{1}{8}$
$\frac{1}{5} = 20\%$	

Q A sum of 1000 on 10% rate of int compounded half yearly figure out the amount?

P = 1000
R = 10% = 5%
T = 2 years $2 \times 2 = 4$

P	A
(20) ²	(2) ²
400	441
$\frac{400}{2} = 200$	$\frac{441}{5} = 88.2$

$441 \times 25 = 11025$
To

Q A sum of 8000 compounded quarterly what will be the compound int after 9 months?

P	A
(20) ³	(2) ³
8000	9261
8000 = 8000	
$\frac{1}{2}$	$\frac{1}{2}$
$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	

P = 8000
R = 20%
T = 9 months

$\frac{20}{4} = 5\%$
(3) $\frac{1}{4} \times 1$
[3 years]

Q P = 10000
R = 10%
T = 2 years 73 days

for 2 years

P	A
(10) ²	(1) ²
100	121
$\frac{1}{2} \times 1000 = 500$	
at year 21	
CI = 2100 - 2 years	
CI = 23100 - 2 years	

